AN4325
Application note

Getting started with STM32F030xx and STM32F070xx series hardware development

Introduction

This application note is intended for system designers who require a hardware implementation overview of the development board features such as the power supply, the clock management, the reset control, the boot mode settings and the debug management. It shows how to use the STM32F0x0xx products family and describes the minimum hardware resources required to develop an application.

This document includes detailed reference design schematics and the description of the main components, interfaces and modes.

Table 1. Applicable products

Type	Part number
Microcontrollers	STM32F030F4, STM32F030CC, STM32F030RC, STM32F030C6, STM32F030K6, STM32F030C8, STM32F030R8, STM32F070C6, STM32F070CB, STM32F070F6, STM32F070RB.

Note: In this document, the notation used for STM32F030xx devices is STM32F030 and the notation used for STM32F070xx devices is STM32F070. When referring to both series the notation STM32F0x0 is used. The pin count and memory size do not impact this hardware description.

AN4325 应用笔记 STM32F030xx 和 STM32F070xx 系列入门

硬件开发

介绍

本应用笔记适用于需要了解开发板功能（例如电源、时钟管理、复位控制、启动模式设置和调试管理）的硬件实现概述的系统设计人员。它展示了如何使用 STM32F0x0xx 产品系列，并描述了开发应用程序所需的最低硬件资源。

本文档包括详细的参考设计原理图以及主要组件、接口和模式的描述。

表 1. 适用产品

Type	Part number
Microcontrollers	STM32F030F4, STM32F030CC, STM32F030RC, STM32F030C6, STM32F030K6, STM32F030C8, STM32F030R8, STM32F070C6, STM32F070CB, STM32F070F6, STM32F070RB.

笔记：在本文档中，STM32F030xx 器件使用的符号为 STM32F030，STM32F070xx 器件使用的符号为 STM32F070。当提及这两个系列时，使用符号 STM32F0x0。引脚数和内存大小不会影响此硬件描述。

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1 Power supplies and reset sources of the STM32F0x0 family

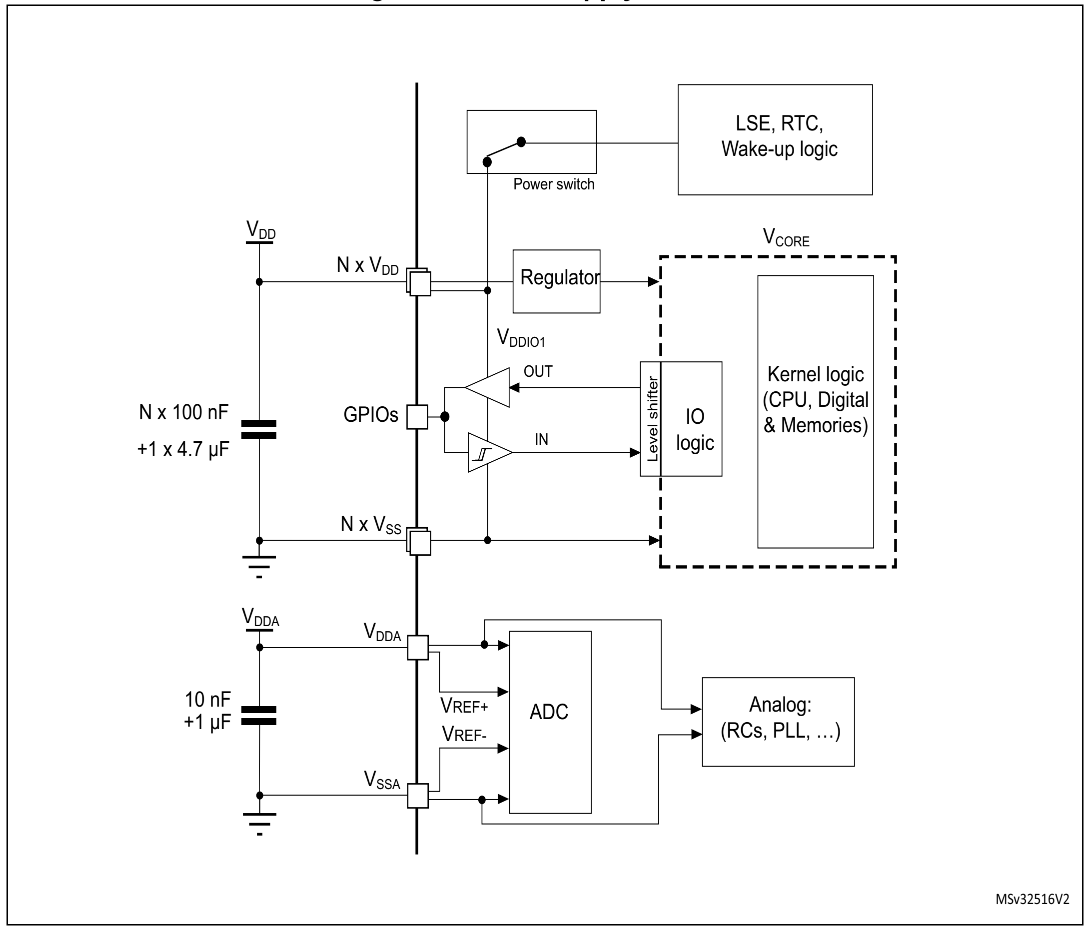
1.1 Power supply schemes

The STM32F0x0 family features different products with different supply schemes. It includes an internal regulator in order to have an internal 1.8 V supply for the core and the digital logic.

There is a variety of power supply schemes:

- V_{DD} from 2.4 V to 3.6 V: external power supply for I/Os and the internal 1.8 V domain. Provided externally through V_{DD} pins.
- V_{DDA} from V_{DD} to 3.6 V: external analog power supply for ADC, Reset blocks, HSI, HSI14, LSI and PLL.
  The V_{DDA} voltage level must always be greater than or equal to the V_{DD} voltage level and must be provided first.

Figure 1. Power supply scheme



1 STM32F0x0 系列的电源和复位源

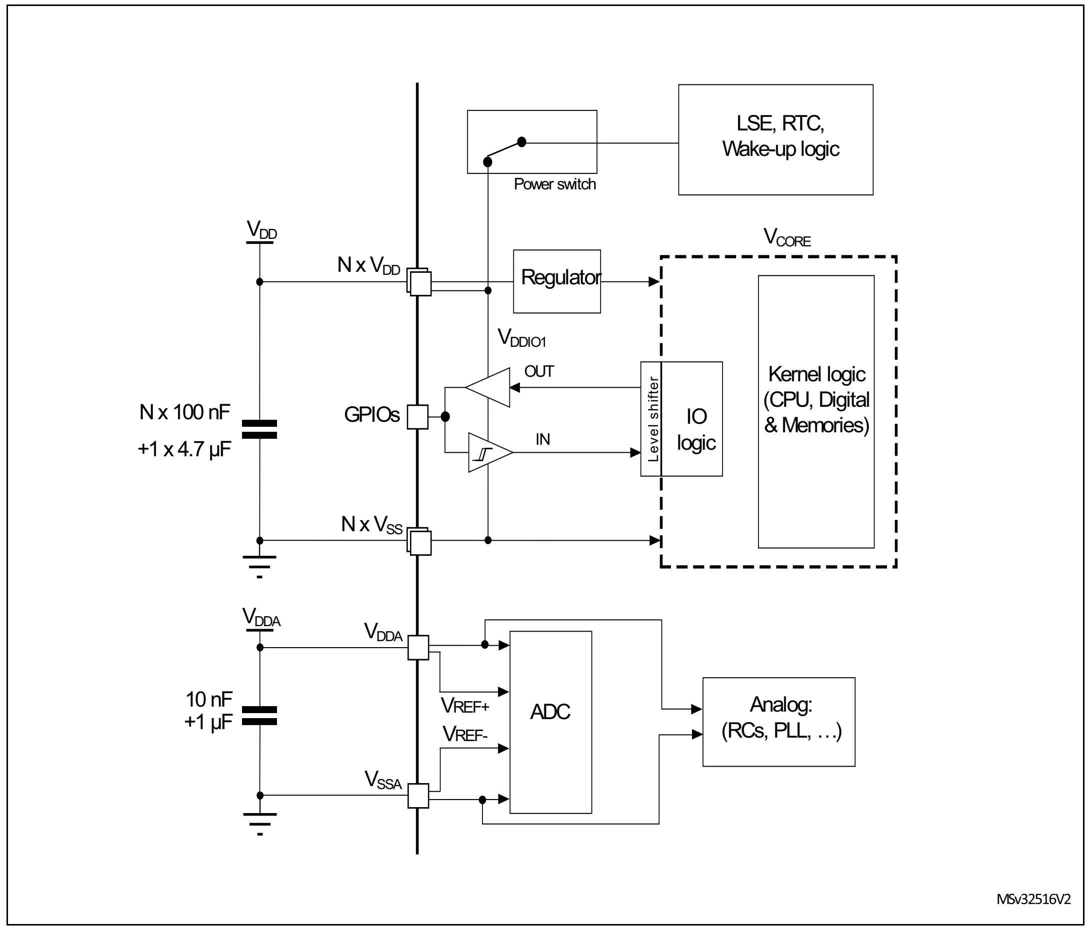
1.1 供电方案

STM32F0x0 系列具有不同的产品和不同的供电方案。它包括一个内部稳压器，以便为内核和数字逻辑提供内部 1.8 V 电源。

有多种供电方案：

- VDD 2.4 V 至 3.6 V：I/O 和内部 1.8 V 域的外部电源。通过 VDD 引脚从外部提供。
- VDDA 从 VDD 到 3.6 V：ADC 、复位模块、 HSI 、 HSI14 、 LSI 和 PLL 的外部模拟电源。 VDDA 电压电平必须始终大于或等于 VDD 电压电平，并且必须首先提供。

图 1. 供电方案



1.1.1 Independent analog power supply

To improve conversion accuracy and to extend the supply flexibility, the analog domain has an independent power supply which can be separately filtered and shielded from noise on the PCB.

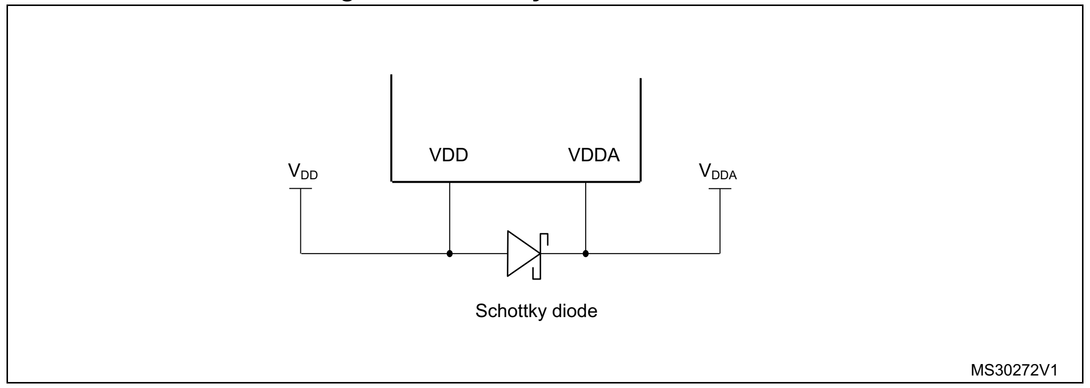
- The ADC voltage supply input is available on a separate VDDA pin.
- An isolated supply ground connection is provided on pin VSSA.

The V_{DDA} supply can be equal to or higher than V_{DD}. This allows V_{DD} to stay low while still providing the full performance for the analog blocks.

When a single supply is used, V_{DDA} must be externally connected to V_{DD}. It is recommended to use an external filtering circuit in order to ensure a noise free V_{DDA}.

When V_{DDA} is different from V_{DD}, V_{DDA} must be always higher or equal to V_{DD}. To keep safe potential difference between V_{DDA} and V_{DD} during power-up/power-down, an external Schottky diode may be used between V_{DD} and V_{DDA}. Refer to the datasheet for the maximum allowed difference.

Figure 2. Schottky diode connection



1.1.2 Voltage regulator

The voltage regulator is always enabled after reset.

It works under two different modes:

- Main (MR) is used in normal operating mode (Run),
- Low power (LPR) can be used in Stop mode where the power demand is reduced.

In standby mode, the regulator is in power-down mode. In this mode, the regulator output is in high impedance and the kernel circuitry is powered down, inducing zero consumption and the loss of the register and SRAM contents. However, the following features are available if configured:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by a hardware option. Once started it cannot be stopped except by a reset.
- Real-time clock (RTC): configured by the RTCEN bit in the RTC domain control register (RCC_BDCR).
- Internal low speed oscillator (LSI): configured by the LSION bit in the Control/Status register (RCC_CSR).
- External 32.768 kHz oscillator (LSE): configured by the LSEON bit in the RTC domain control register (RCC_BDCR).

1.1.1 独立模拟电源

为了提高转换精度并扩展供电灵活性，模拟域具有独立的电源，可以单独过滤并屏蔽 PCB 上的噪声。

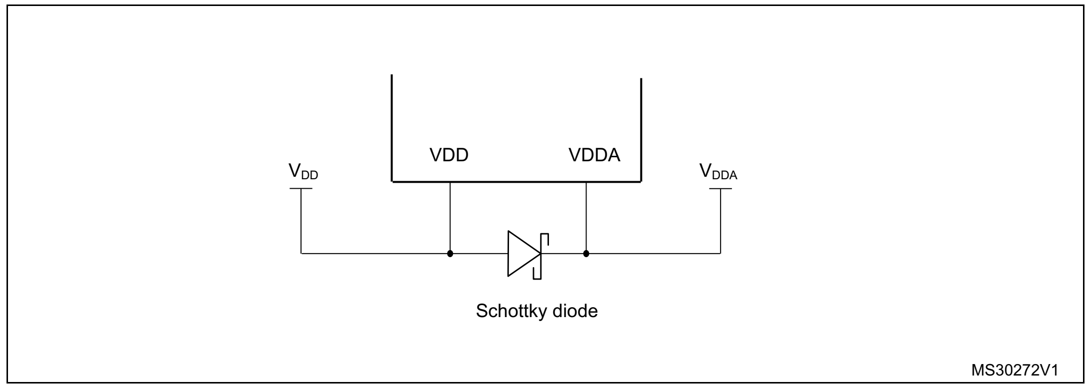
- ADC 电源电压输入可在单独的 VDDA 引脚上使用。
- 引脚 VSSA 上提供隔离电源接地连接。

VDDA 电源可以等于或高于 VDD。这使得 VDD 保持在低水平，同时仍为模拟模块提供全部性能。

当使用单电源时，VDDA 必须外部连接到 VDD。建议使用外部滤波电路以确保 VDDA 无噪声。

当 VDDA 与 VDD 不同时，VDDA 必须始终高于或等于 VDD。为了在上电 / 断电期间保持 VDDA 和 VDD 之间的安全电位差，可以在 VDD 和 VDDA 之间使用外部肖特基二极管。有关最大允许差异，请参阅数据表。

图 2. 肖特基二极管连接



1.1.2 电压调节器

电压调节器在复位后始终启用。

它在两种不同的模式下工作：

- Main （MR）用于正常运行模式（Run），
- 低功耗 (LPR) 可用于降低功耗需求的停止模式。

在待机模式下，稳压器处于断电模式。在此模式下，调节器输出处于高阻抗，内核电路断电，导致零功耗以及寄存器和 SRAM 内容的丢失。但是，如果配置，则可以使用以下功能：

- 独立看门狗（IWDG）：IWDG 通过写入其 Key 寄存器或通过硬件选项来启动。一旦启动，除非重置，否则无法停止。
- 实时时钟（RTC）：由 RTC 域控制寄存器（RCC_BDCR）中的 RTCEN 位配置。
- 内部低速振荡器（LSI）：由控制 / 状态寄存器（RCC_CSR）中的 LSION 位配置。
- 外部 32.768 kHz 振荡器（LSE）：由 RTC 域控制寄存器（RCC_BDCR）中的 LSEON 位配置。

1.2 Reset and power supply supervisor

1.2.1 Power-on reset (POR) / power-down reset (PDR)

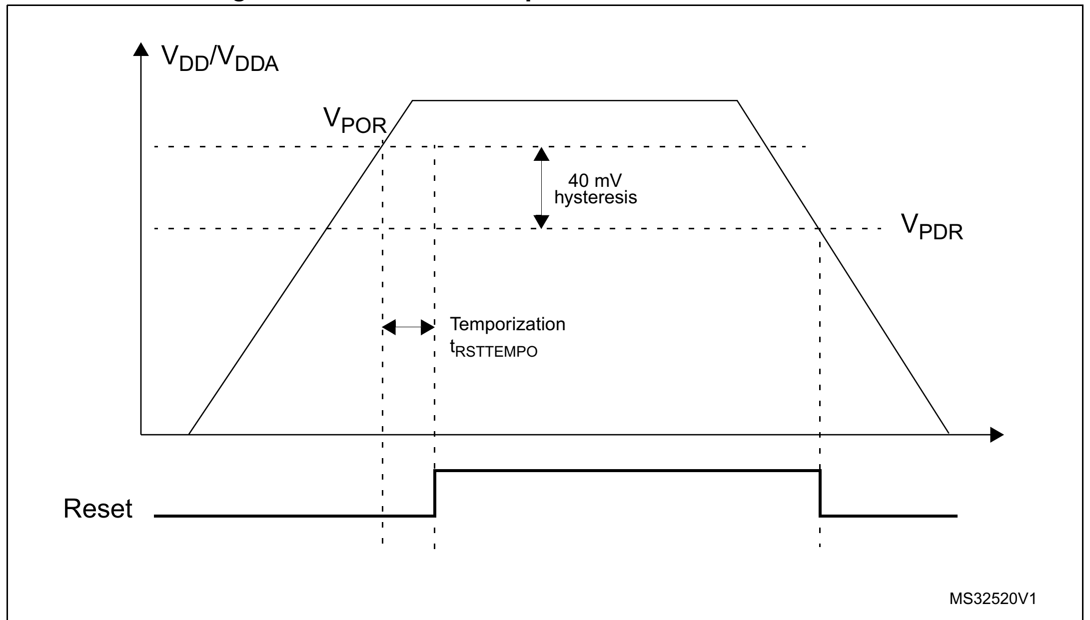
The device has an integrated power-on reset (POR) and power-down reset (PDR) circuits which are always active and ensure proper operation above a threshold of 2.4 V.

The device remains in Reset mode when the monitored supply voltage is below a specified threshold, $V_{POR/PDR}$, without the need for an external reset circuit.

- The POR monitors only the V_{DD} supply voltage. During the startup phase, V_{DDA} must arrive first and be greater than or equal to $V_{DD.}$
- The PDR monitors both the V_{DD} and V_{DDA} supply voltages. However, the V_{DDA} power supply supervisor can be disabled (by programming a dedicated option bit $V_{DDA\_MONITOR}$) to reduce the power consumption if the application design ensures that V_{DDA} is higher than or equal to V_{DD}.

For more details on the power on / power down reset threshold, refer to the electrical characteristics section in the datasheet.

Figure 3. Power on reset/power down reset waveform



1.2 复位和电源监控器

1.2.1 上电复位（ POR ） / 掉电复位（ PDR ）

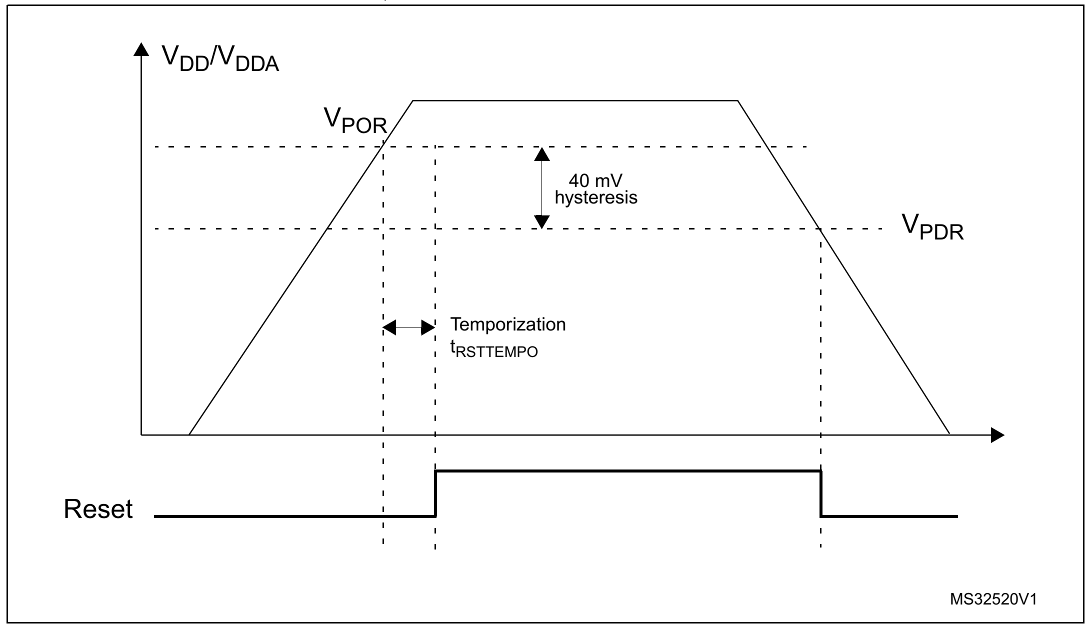
该器件具有集成的上电复位 (POR) 和断电复位 (PDR) 电路，这些电路始终处于活动状态并确保在高于 2.4 V 阈值时正常工作。

当监控的电源电压低于指定阈值 VPOR/PDR 时，器件保持复位模式，无需外部复位电路。

- POR 仅监控 VDD 电源电压。在启动阶段，VDDA 必须首先到达并且大于或等于 VDD。
- PDR 监控 VDD 和 VDDA 电源电压。然而，如果应用设计确保 VDDA 高于或等于 VDD，则可以禁用 VDDA 电源监控器（通过编程专用选项位 VDDA_MONITOR ）以降低功耗。

有关加电 / 断电复位阈值的更多详细信息，请参阅数据表中的电气特性部分。

图 3. 上电复位 / 掉电复位波形



1.2.2 System reset

A system reset sets all registers to their reset values, except the reset flags in the clock controller CSR register and the registers in the RTC domain. A system reset is generated when one of the following events occurs:

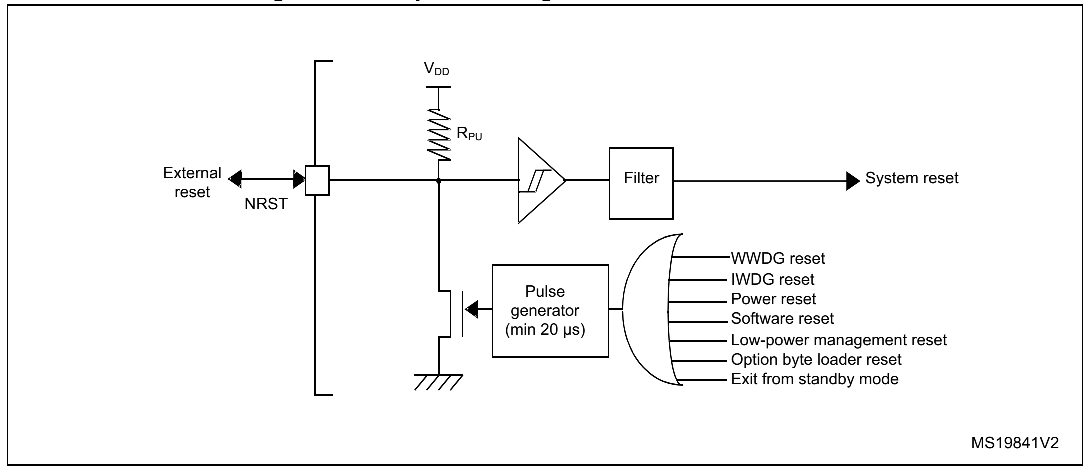
- A low level on the NRST pin (external reset)
- System window watchdog event (WWDG reset)
- Independent watchdog event (IWDG reset)
- A software reset (SW reset)
- Low-power management reset
- Option byte loader reset
- Power reset.

The reset source can be identified by checking the reset flags in the Control/Status register, RCC_CSR.

The RESET service routine vector is fixed at address 0x0000_0004 in the memory map.

The system reset signal provided to the device is output on the NRST pin. The pulse generator guarantees a minimum reset pulse duration of 20 µs for each internal reset source. In the case of an external reset, the reset is generated while the NRST pin is asserted low.

Figure 4. Simplified diagram of the reset circuit



Software reset

The SYSRESETREQ bit in Cortex-M0 Application Interrupt and Reset Control Register must be set to force a software reset on the device. Refer to the Cortex®-M0 technical reference manual for more details.

1.2.2 系统复位

系统复位将所有寄存器设置为其复位值，时钟控制器 CSR 寄存器中的复位标志和 RTC 域中的寄存器除外。当发生以下事件之一时，会生成系统重置：

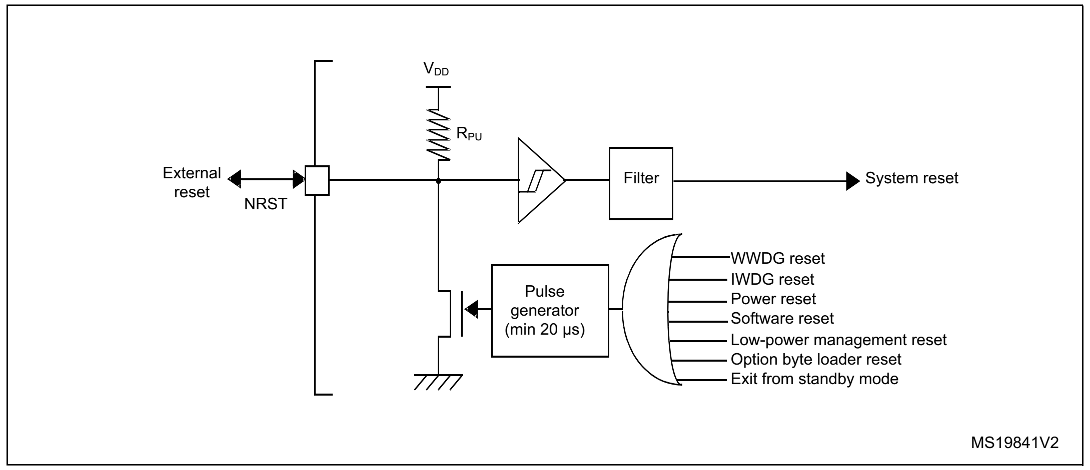
- NRST 引脚上的低电平（外部复位）
- 系统窗口看门狗事件（ WWDG 重置）
- 独立看门狗事件（ IWDG 复位）
- 软件复位（ SW 复位）
- 低功耗管理重置
- 选项字节加载器重置
- 电源重置。

通过检查控制 / 状态寄存器 RCC_CSR 中的复位标志可以识别复位源。

RESET 服务例程向量固定在内存映射中的地址 0x0000_0004 处。

提供给器件的系统复位信号在 NRST 引脚上输出。脉冲发生器保证每个内部复位源的最小复位脉冲持续时间为 20 µs。在外部复位的情况下，当 NRST 引脚被置为低电平时会产生复位。

图 4. 复位电路的简化图



软件复位

Cortex-M0 应用中断和复位控制寄存器中的 SYSRESETREQ 位必须置位，以强制对器件进行软件复位。有关更多详细信息，请参阅 Cortex®-M0 技术参考手册。

Low-power mode security reset

To prevent that critical applications mistakenly enter a low-power mode, two low-power mode security resets are available. If enabled in Option bytes, the resets are generated in the following conditions:

- Entering Standby mode: This type of reset is enabled by resetting nRST_STDBY bit in User Option Bytes. In this case, whenever a Standby mode entry sequence is successfully executed, the device is reset instead of entering Standby mode.
- Entering Stop mode: This type of reset is enabled by resetting nRST_STOP bit in User Option Bytes. In this case, whenever a Stop mode entry sequence is successfully executed, the device is reset instead of entering Stop mode.

Option byte loader reset

The option byte loader reset is generated when OBL_LAUNCH (bit 13) is set in the FLASH_CR register. This bit launches the option byte loading by software.

Power reset

A power reset sets all registers to their reset values, except the RTC domain. See *Table 2*.

RTC domain reset

An RTC domain reset only affects the RTC, LSE and LSI. It is generated when one of the following events occurs. See*Table 2*.

Table 2. System reset

Modes	Power reset	RTC domain reset
POR/PDR reset	yes	yes
Exiting Standby mode	yes	no
Setting the BDRST bit in RCC_BDCR	no	yes

低功耗模式安全重置

为了防止关键应用程序错误地进入低功耗模式，可以使用两种低功耗模式安全重置。如果在选项字节中启用，则在以下条件下会生成复位：

- 进入待机模式：通过复位用户选项字节中的 nRST_STDBY 位来启用此类复位。在这种情况下，只要成功执行待机模式进入序列，设备就会复位，而不是进入待机模式。
- 进入停止模式：通过复位用户选项字节中的 nRST_STOP 位来启用此类复位。在这种情况下，只要成功执行停止模式进入序列，器件就会复位，而不是进入停止模式。

选项字节加载器复位

当 FLASH_CR 寄存器中的 OBL_LAUNCH （位 13 ）置位时，会生成选项字节加载器复位。该位启动软件加载选项字节。

电源复位

电源复位将所有寄存器设置为其复位值（ RTC 域除外）。见表 2 .

RTC 域复位

RTC 域复位仅影响 RTC 、 LSE 和 LSI。当发生以下事件之一时会生成它。参见表 2。

表 2. 系统重置

Modes	Power reset	RTC domain reset
POR/PDR reset	yes	yes
Exiting Standby mode	yes	no
Setting the BDRST bit in RCC_BDCR	no	yes

2 Clocks

Different clock sources can be used to drive the system clock (SYSCLK):

- HSI 8 MHz RC oscillator clock (high-speed internal clock signal)
- HSE oscillator clock (high-speed external clock signal)
- PLL clock

The devices have other secondary clock sources:

- 40 kHz low-speed internal RC (LSI RC) that drives the independent watchdog and, optionally, the RTC used for auto-wakeup from the Stop/Standby modes.
- 32.768 kHz low-speed external crystal (LSE crystal) that optionally drives the RTC.
- HSI 14MHz RC oscillator (HSI14) dedicated for ADC.

Each clock source can be switched on or off independently when it is not used, to optimize the power consumption. Refer to reference manual *STM32F030x4/6/8/C and STM32F070x6/B advanced ARM®-based 32-bit MCUs* (RM0360) for a description of the clock tree.

2 个时钟

可以使用不同的时钟源来驱动系统时钟（ SYSCLK ）：

- HSI 8 MHz RC 振荡器时钟（高速内部时钟信号）
- HSE 振荡器时钟（高速外部时钟信号）
- PLL 时钟

这些设备还有其他辅助时钟源：

- 40 kHz 低速内部 RC (LSI RC)，用于驱动独立看门狗以及用于从停止 / 待机模式自动唤醒的 RTC （可选）。
- 32.768 kHz 低速外部晶振（ LSE 晶振），可选择驱动 RTC。
- HSI 14MHz RC 振荡器（ HSI14 ）专用于 ADC。

每个时钟源在不使用时可以独立打开或关闭，以优化功耗。有关时钟树的说明，请参阅参考手册 STM32F030x4/6/8/C 和 STM32F070x6/B 高级基于 ARM® 的 32 位 MCU (RM0360)。

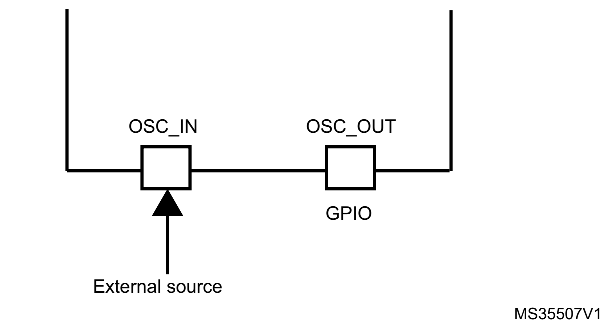
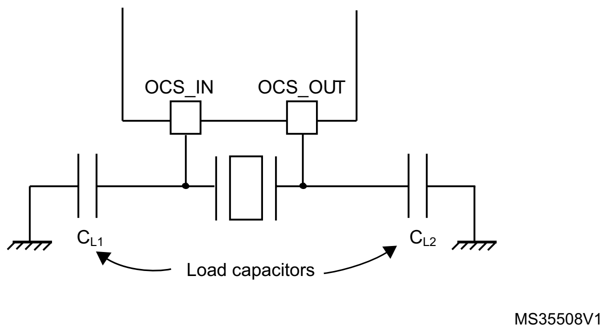
2.1 High speed external clock signal (HSE) OSC clock

The high speed external clock signal can be generated from two possible clock sources:

- HSE external crystal/ceramic resonator
- HSE user external clock.

The resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and the startup stabilization time. The loading capacitance values must be adjusted according to the selected resonator.

Figure 5. HSE/ LSE clock sources

<table>
<tr><th>Clock source</th><th>Hardware configuration</th></tr>
<tr><td>External clock</td><td>
</td></tr>
<tr><td>Crystal/Ceramic resonators</td><td>
</td></tr>
</table>

External crystal/ceramic resonator (HSE crystal)

The 4 to 32 MHz external oscillator has the advantage of producing a very accurate frequency on the main clock. Refer to the electrical characteristics section of the datasheet for more details about the associated hardware configuration.

The HSERDY flag in the Clock control register (RCC_CR) indicates if the HSE oscillator is stable or not. At startup, the clock is not released until this bit is set by hardware. An interrupt can be generated if enabled in the Clock interrupt register (RCC_CIR).

The HSE Crystal can be switched on and off using the HSEON bit in the Clock control register (RCC_CR).

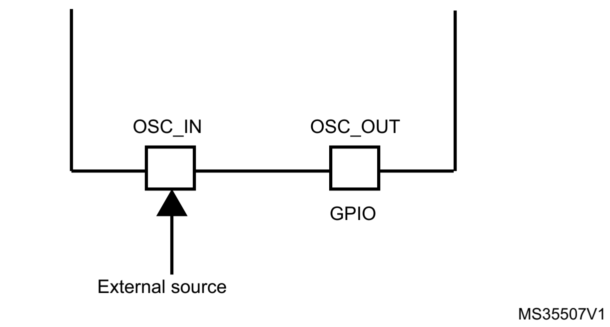
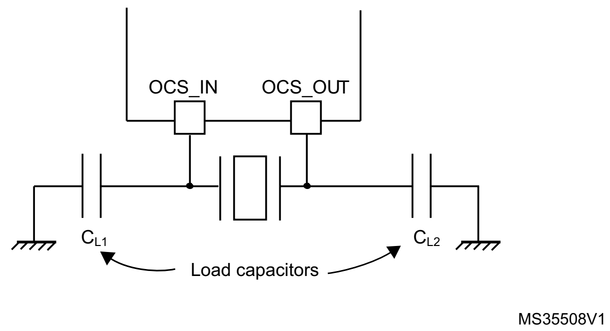
2.1 高速外部时钟信号（HSE） OSC 时钟

高速外部时钟信号可以从两个可能的时钟源生成：

- HSE 外部晶体 / 陶瓷谐振器
- HSE 用户外部时钟。

谐振器和负载电容器必须尽可能靠近振荡器引脚放置，以最大限度地减少输出失真和启动稳定时间。负载电容值必须根据所选谐振器进行调整。

图 5. HSE/LSE 时钟源

<table>
<tr><th>Clock source</th><th>Hardware configuration</th></tr>
<tr><td>External clock</td><td>
</td></tr>
<tr><td>Crystal/Ceramic resonators</td><td>
</td></tr>
</table>

外部晶体 / 陶瓷谐振器（HSE 晶体）

4 至 32 MHz 外部振荡器的优点是可以在主时钟上产生非常精确的频率。有关相关硬件配置的更多详细信息，请参阅数据表的电气特性部分。

时钟控制寄存器（RCC_CR）中的 HSERDY 标志指示 HSE 振荡器是否稳定。启动时，只有硬件设置该位后，时钟才会释放。如果在时钟中断寄存器（RCC_CIR）中使能，则可以生成中断。

HSE 晶振可以使用时钟控制寄存器 (RCC_CR) 中的 HSEON 位打开和关闭。

External source (HSE bypass)

In this mode, an external clock source must be provided. It can have a frequency of up to 32 MHz. The user selects this mode by setting the HSEBYP and HSEON bits in the *Clock control register (RCC_CR)*. The external clock signal (square, sinus or triangle) with ~40-60% duty cycle depending on the frequency (refer to the datasheet) has to drive the OSC_IN pin while the OSC_OUT pin can be used a GPIO. See *Figure 5*.

2.2 LSE clock

The LSE crystal is a 32.768 kHz Low Speed External crystal or ceramic resonator. It has the advantage of providing a low-power but highly accurate clock source to the real-time clock peripheral (RTC) for clock/calendar or other timing functions.

The LSE crystal is switched on and off using the LSEON bit in RTC domain control register (RCC_BDCR). The crystal oscillator driving strength can be changed at runtime using the LSEDRV[1:0] bits in the RTC domain control register (RCC_BDCR) to obtain the best compromise between robustness and short start-up time on one side and low power-consumption on the other.

The LSERDY flag in the RTC domain control register (RCC_BDCR) indicates whether the LSE crystal is stable or not. At startup, the LSE crystal output clock signal is not released until this bit is set by hardware. An interrupt can be generated if enabled in the Clock interrupt register (RCC_CIR).

External source (LSE bypass)

In this mode, an external clock source must be provided. It can have a frequency of up to 1 MHz. The user selects this mode by setting the LSEBYP and LSEON bits in the RTC domain control register (RCC_BDCR). The external clock signal (square, sinus or triangle) has to drive the OSC32_IN pin while the OSC32_OUT pin can be used as GPIO. See *Figure 5*.

2.3 HSI clock

The HSI clock signal is generated from an internal 8 MHz RC oscillator and can be used directly as a system clock or divided by 2 to be used as PLL input. The HSI RC oscillator has the advantage of providing a clock source at low cost (no external components). It also has a faster startup time than the HSE crystal oscillator however, even with calibration, the frequency is less accurate than an external crystal oscillator or ceramic resonator.

Calibration

The RC oscillator frequencies can vary from one chip to another due to manufacturing process variations, Therefore, it is possible to route the HSI clock to the MCO multiplexer, then the clock can be input to Timer 14 allowing the user to calibrate the oscillator.

2.4 LSI clock

The LSI RC acts as an low-power clock source that can be kept running in Stop and Standby mode for the independent watchdog (IWDG) and RTC. The clock frequency is

外部源（HSE 旁路）

在此模式下，必须提供外部时钟源。它的频率最高可达 32 MHz。用户通过设置时钟控制寄存器（RCC_CR）中的 HSEBYP 和 HSEON 位来选择该模式。占空比为 60% （取决于频率）的外部时钟信号（方波、正弦波或三角波）必须驱动 OSC_IN 引脚，而 OSC _OUT 引脚可用作 GPIO。参见图 5。

2.2 伦敦证券交易所时钟

LSE 晶体是 32.768 kHz 低速外部晶体或陶瓷谐振器。它的优点是为实时时钟外设（RTC）提供低功耗但高精度的时钟源，以实现时钟 / 日历或其他计时功能。

LSE 晶振使用 RTC 域控制寄存器 (RCC_BDCR) 中的 LSEON 位来打开和关闭。可以使用 RTC 域控制寄存器 (RCC_BDCR) 中的 LSEDRV[1:0] 位在运行时更改晶体振荡器驱动强度，以获得鲁棒性和短启动时间与低功耗之间的最佳折衷。

RTC 域控制寄存器（RCC_BDCR）中的 LSERDY 标志指示 LSE 晶振是否稳定。启动时，LSE 晶振输出时钟信号只有在硬件设置该位后才会释放。如果在时钟中断寄存器（RCC _CIR）中使能，则可以生成中断。

外部源（LSE 旁路）

在此模式下，必须提供外部时钟源。它的频率最高可达 1 MHz。用户通过设置 RTC 域控制寄存器（RCC_BDCR）中的 LSEBYP 和 LSEON 位来选择该模式。外部时钟信号（方波、正弦波或三角波）必须驱动 OSC32_IN 引脚，而 OSC32_OUT 引脚可用作 GPIO。参见图 5。

2.3 恒指时钟

HSI 时钟信号由内部 8 MHz RC 振荡器产生，可直接用作系统时钟或除以 2 作为 PLL 输入。HSI RC 振荡器的优点是能够以低成本提供时钟源（无需外部元件）。它的启动时间也比 HSE 晶体振荡器更快，但即使经过校准，频率也不如外部晶体振荡器或陶瓷谐振器准确。

校准

由于制造工艺的差异，不同芯片的 RC 振荡器频率可能有所不同，因此，可以将 HSI 时钟路由到 MCO 多路复用器，然后将时钟输入到定时器 14，从而允许用户校准振荡器。

2.4LSI 时钟

LSI RC 充当低功耗时钟源，可在独立看门狗 (IWDG) 和 RTC 的停止和待机模式下保持运行。时钟频率为

around 40 kHz (between 30 kHz and 60 kHz). For more details, refer to the electrical characteristics section of the datasheets.

2.5 ADC clock

The ADC clock is either the dedicated 14 MHz RC oscillator (HSI14) or PCLK divided by 2 or 4. When the ADC clock is derived from PCLK, it is in an opposite phase with PCLK. The 14 MHz RC oscillator can be configured by software either to be turned on/off ("auto-off mode") by the ADC interface or to be always enabled.

2.6 Clock security system (CSS)

The clock security system can be activated by software. In this case, the clock detector is enabled after the HSE oscillator startup delay, and disabled when this oscillator is stopped.

- If a failure is detected on the HSE oscillator clock, the oscillator is automatically disabled.
  - A clock failure event is sent to the break inputs of TIM1 advanced control timer and TIM15, TIM16 and TIM17 general purpose timers.
  - An interrupt is generated to inform the software about the failure (clock security system interrupt CSSI), allowing the MCU to perform recovery operations.
  - CSSI is linked to the Cortex®-M0 NMI (non-maskable interrupt) exception vector.
- If the HSE oscillator is used directly or indirectly as the system clock (indirectly means that it is used as the PLL input clock, and the PLL clock is used as the system clock), a detected failure causes a switch of the system clock to the HSI oscillator and the disabling of the external HSE oscillator. If the HSE oscillator clock (divided or not) is the clock entry of the PLL that is being used as a system clock when the failure occurs, the PLL is disabled too.

For details, see the reference manual *STM32F030x4/6/8/C and STM32F070x6/B advanced ARM®-based 32-bit MCUs* (RM0360) available from the STMicroelectronics website www.st.com.

约 40 kHz （30 kHz 至 60 kHz 之间）。有关更多详细信息，请参阅数据表的电气特性部分。

2.5 ADC 时钟

ADC 时钟是专用 14 MHz RC 振荡器 (HSI14) 或除以 2 或 4 的 PCLK。当 ADC 时钟源自 PCLK 时，它与 PCLK 相位相反。 14 MHz RC 振荡器可通过软件配置为通过 ADC 接口打开 / 关闭（“自动关闭模式”）或始终启用。

2.6 时钟安全系统（CSS）

时钟安全系统可以通过软件激活。在这种情况下，时钟检测器在 HSE 振荡器启动延迟后启用，并在该振荡器停止时禁用。

- 如果检测到 HSE 振荡器时钟出现故障，振荡器将自动禁用。 – 时钟故障事件发送至 TIM1 高级控制定时器以及 TIM15 、 TIM16 和 TIM17 通用定时器的中断输入。 – 生成中断以通知软件有关故障（时钟安全系统中断 CSSI），从而允许 MCU 执行恢复操作。 – CSSI 链接到 Cortex®-M0 NMI （不可屏蔽中断）异常向量。

- 如果直接或间接使用 HSE 振荡器作为系统时钟（间接意味着它用作 PLL 输入时钟，并且 PLL 时钟用作系统时钟），检测到的故障会导致系统时钟切换到 HSI 振荡器并禁用外部 HSE 振荡器。如果 HSE 振荡器时钟（分频或未分频）是发生故障时用作系统时钟的 PLL 的时钟条目，PLL 也被禁用。

有关详细信息，请参阅意法半导体网站 www.st.com 上提供的参考手册 STM32F030x4/6/8/C 和 STM32F070x6/B 高级基于 ARM® 的 32 位 MCU (RM0360)。

3 Boot configuration

In the STM32F0x0, three different boot modes can be selected through the BOOT0 pin and boot configuration bits nBOOT1 in the User option byte, as shown in the following table:

Table 3. Boot modes

Boot mode configuration		Mode
nBOOT1 bit	BOOT0 pin	
x	0	**Main Flash memory** is selected as boot space(1)
1	1	**System memory** is selected as boot space
0	1	**Embedded SRAM** is selected as boot space

1. For STM32F070x6 and STM32F030xC devices, see also Empty check description.

The Boot mode configuration is latched on the 4th rising edge of SYSCLK after a reset. It is up to the user to set Boot mode configuration related to the required Boot mode.

The Boot mode configuration is also re-sampled when exiting from Standby mode. Consequently they must be kept in the required Boot mode configuration in Standby mode. After this startup delay has elapsed, the CPU fetches the top-of-stack value from address 0x0000 0000, then starts code execution from the boot memory at 0x0000 0004.

Depending on the selected boot mode, main Flash memory, system memory or SRAM is accessible as follows:

- Boot from main Flash memory: the main Flash memory is aliased in the boot memory space (0x0000 0000), but still accessible from its original memory space (0x0800 0000). In other words, the Flash memory contents can be accessed starting from address 0x0000 0000 or 0x0800 0000.
- Boot from system memory: the system memory is aliased in the boot memory space (0x0000 0000), but still accessible from its original memory space (0x1FFF EC00 on STM32F030x4, STM32F030x6 and STM32F030x8 devices, 0x1FFF C400 on STM32F070x6 devices, 0x1FFF C800 on STM32F070xB and 0x1FFF D800 on STM32F030xC devices).
- Boot from the embedded SRAM: the SRAM is aliased in the boot memory space (0x0000 0000), but it is still accessible from its original memory space (0x2000 0000).

Empty check

On STM32F070x6 and STM32F030xC devices only, internal empty check flag is implemented to allow easy programming of the virgin devices by the boot loader. This flag is used when BOOT0 pin is defining Main Flash memory as the target boot space. When the flag is set, the device is considered as empty and System memory (boot loader) is selected instead of the Main Flash as a boot space to allow user to program the Flash memory.

This flag is updated only during Option bytes loading: it is set when the content of the address 0x08000 0000 is read as 0xFFFF FFFF, otherwise it is cleared. It means a power on or setting of OBL_LAUNCH bit in FLASH_CR register is needed to clear this flag after programming of a virgin device to execute user code after System reset.

3 启动配置

在 STM32F0x0 中，可以通过 BOOT0 引脚和用户选项字节中的启动配置位 nBOOT1 来选择三种不同的启动模式，如下表所示：

表 3. 引导模式

Boot mode configuration		Mode
nBOOT1 bit	BOOT0 pin	
x	0	**Main Flash memory** is selected as boot space(1)
1	1	**System memory** is selected as boot space
0	1	**Embedded SRAM** is selected as boot space

1. For STM32F070x6 and STM32F030xC devices, see also Empty check description.

复位后，引导模式配置在 SYSCLK 的第 4 个上升沿被锁存。由用户设置与所需启动模式相关的启动模式配置。

当退出待机模式时，启动模式配置也会重新采样。因此，它们必须在待机模式下保持所需的引导模式配置。经过此启动延迟后，CPU 从地址 0x0000 0000 获取堆栈顶部值，然后从 0x0000 0004 处的启动存储器开始执行代码。

根据所选的启动模式，可以按如下方式访问主闪存、系统内存或 SRAM：

- 从主闪存启动：主闪存在启动存储空间（0x0000 0000）中具有别名，但仍可从其原始存储空间（0x0800 0000）访问。换句话说，可以从地址 0x0000 0000 或 0x0800 0000 开始访问 Flash 存储器内容。
- 从系统内存启动：系统内存在启动内存空间中使用别名（0x0000 0000），但仍可从其原始内存空间访问（STM32F030x4、STM32F030x6 和 STM32F030x8 器件上为 0x1FFF EC00，STM32F070x6 器件上为 0x1FFF C400，STM32F070xB 器件上为 0x1FFF C800，STM32F030xC 器件上为 0x1FFF D800）。
- 从嵌入式 SRAM 启动：SRAM 在启动存储空间（0x0000 0000）中是别名的，但仍然可以从其原始存储空间（0x2000 0000）访问。

空支票

仅在 STM32F070x6 和 STM32F030xC 器件上，实现了内部空检查标志，以允许引导加载程序轻松对原始器件进行编程。当 BOOT0 引脚将主闪存定义为目标引导空间时，使用该标志。当设置该标志时，设备被视为空，并且选择系统存储器（引导加载程序）而不是主闪存作为引导空间，以允许用户对闪存进行编程。

该标志仅在选项字节加载期间更新：当地址 0x08000 0000 的内容读取为 0xFFFF FFFF 时设置该标志，否则清除。这意味着在系统复位后对原始器件进行编程以执行用户代码后，需要上电或设置 FLASH_CR 寄存器中的 OBL_LAUNCH 位来清除该标志。

Note: If the device is programmed for a first time but the Option bytes are not reloaded, the device will still select System memory as a boot space after a System reset. The boot loader code is able to detect this situation and will change the boot memory mapping to Main Flash and perform a jump to user code programmed there.

Physical remap

Once the boot mode is selected, the application software can modify the memory accessible in the code area. This modification is performed by programming the MEM_MODE bits in the SYSCFG configuration register 1 (SYSCFG_CFGR1). Unlike Cortex® M3 and M4, the M0 CPU does not support the vector table relocation. For application code which is located in a different address than 0x0800 0000, some additional code must be added in order to be able to serve the application interrupts. A solution will be to relocate by software the vector table to the internal SRAM:

- Copy the vector table from the Flash (mapped at the base of the application load address) to the base address of the SRAM at 0x2000 0000.
- Remap SRAM at address 0x0000 0000, using SYSCFG configuration register 1.
- Then once an interrupt occurs, the Cortex®-M0 processor will fetch the interrupt handler start address from the relocated vector table in SRAM, then it will jump to execute the interrupt handler located in the Flash.

This operation should be done at the initialization phase of the application. Please refer to application note *STM32F0xx in-application programming using the USART* (AN4065) and attached IAP code from *www.st.com* for more details.

Embedded boot loader

The embedded boot loader is located in the System memory, programmed by ST during production. It is used to reprogram the Flash memory using one of the following serial interfaces:

- USART on pins PA14/PA15 or PA9/PA10
- I2C on pins PB6/PB7 (STM32F070xx and STM32F030xC devices only)
- USB DFU interface (STM32F070xx devices only)

For further details, please refer to application note *STM32 microcontroller system memory boot mode* (AN2606).

笔记： 如果首次对器件进行编程但未重新加载选项字节，则器件在系统复位后仍会选择系统内存作为引导空间。引导加载程序代码能够检测到这种情况，并将更改引导存储器映射到主闪存并执行跳转到在那里编程的用户代码。

物理重映射

一旦选择了启动模式，应用软件就可以修改代码区域中可访问的存储器。此修改是通过对 SYSCFG 配置寄存器 1 (SYSCFG_CFGR1) 中的 MEM_MODE 位进行编程来执行的。与 Cortex® M3 和 M4 不同，M0 CPU 不支持向量表重定位。对于位于与 0x0800 0000 不同的地址的应用程序代码，必须添加一些附加代码以便能够为应用程序中断提供服务。解决方案是通过软件将向量表重新定位到内部 SRAM：

- 将向量表从闪存（映射到应用程序加载地址的基址）复制到 SRAM 的基址 0x2000 0000。
- 使用 SYSCFG 配置寄存器 1 重新映射地址 0x0000 0000 处的 SRAM。
- 一旦中断发生，Cortex®-M0 处理器将从 SRAM 中重定位的向量表中取出中断处理程序起始地址，然后跳转到执行位于 Flash 中的中断处理程序。

此操作应在应用程序的初始化阶段完成。请参阅应用笔记 STM32F0xx 使用 USART (AN4065) 进行应用编程以及来自 www.st.com 的附加 IAP 代码，了解更多详细信息。

嵌入式引导加载程序

嵌入式引导加载程序位于系统存储器中，由 ST 在生产过程中进行编程。它用于使用以下串行接口之一对闪存重新编程：

- USART 位于引脚 PA14/PA15 或 PA9/PA10 上
- 引脚 PB6/PB7 上的 I2C （仅限 STM32F070xx 和 STM32F030xC 器件）
- USB DFU 接口（仅限 STM32F070xx 器件）

如需了解更多详细信息，请参阅应用笔记 STM32 微控制器系统存储器启动模式 (AN2606)。

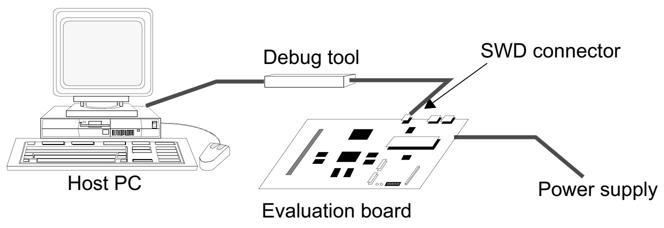
4 Debug management

4.1 Introduction

The host/target interface is the hardware equipment that connects the host to the application board. This interface is made of three components: a hardware debug tool, an SWD connector and a cable connecting the host to the debug tool.

Figure 6 shows the connection of the host to the evaluation board.

Figure 6. Host-to-board connection



4.2 SWD port (serial wire debug)

The STM32F0x0 core integrates the serial wire debug port (SW-DP). It is an ARM® standard CoreSight™ debug port with a 2-pin (clock + data) interface to the debug access port.

4.3 Pinout and debug port pins

The STM32F0x0 MCU is offered in various packages with varying numbers of available pins.

4.3.1 Serial wire debug (SWD) pin assignment

The same SWD pin assignment is available on all STM32F0x0 packages.

Table 4. SWD port pins

SWD pin name	SWD port		Pin assignment
	Type	Debug assignment	
SWDIO	I/O	Serial wire data input/output	PA13
SWCLK	I	Serial wire clock	PA14

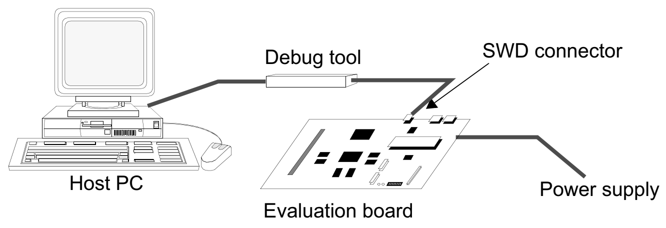
4 调试管理

4.1 简介

主机 / 目标接口是将主机连接到应用板的硬件设备。该接口由三个组件组成：硬件调试工具、SWD 连接器以及连接主机和调试工具的电缆。

图 6 显示了主机与评估板的连接。

图 6. 主机到板连接



4.2 SWD 口（串口线调试）

STM32F0x0 内核集成了串行线调试端口 (SW-DP)。它是一个 ARM®标准 CoreSight™ 调试端口，具有连接调试访问端口的 2 引脚（时钟 + 数据）接口。

4.3 引脚分配和调试端口引脚

STM32F0x0 MCU 采用各种封装，具有不同数量的可用引脚。

4.3.1 串行线调试 (SWD) 引脚分配

所有 STM32F0x0 封装上都具有相同的 SWD 引脚分配。

表 4. SWD 端口引脚

SWD pin name	SWD port		Pin assignment
	Type	Debug assignment	
SWDIO	I/O	Serial wire data input/output	PA13
SWCLK	I	Serial wire clock	PA14

4.3.2 SWD pin assignment

After reset (SYSRESETn or PORESETn), the pins used for the SWD are assigned as dedicated pins which are immediately usable by the debugger host.

However, the MCU offers the possibility to disable the SWD, therefore releasing the associated pins for general-purpose I/O (GPIO) usage. For more details on how to disable SWD port, check reference manual *STM32F030x4/6/8/C and STM32F070x6/B advanced ARM®-based 32-bit MCUs* (RM0360) on the section of I/O pin alternate function multiplexer and mapping.

4.3.3 Internal pull-up and pull-down on SWD pins

Once the SWD I/O is released by the user software, the GPIO controller takes control of these pins. The reset states of the GPIO control registers put the I/Os in the equivalent states:

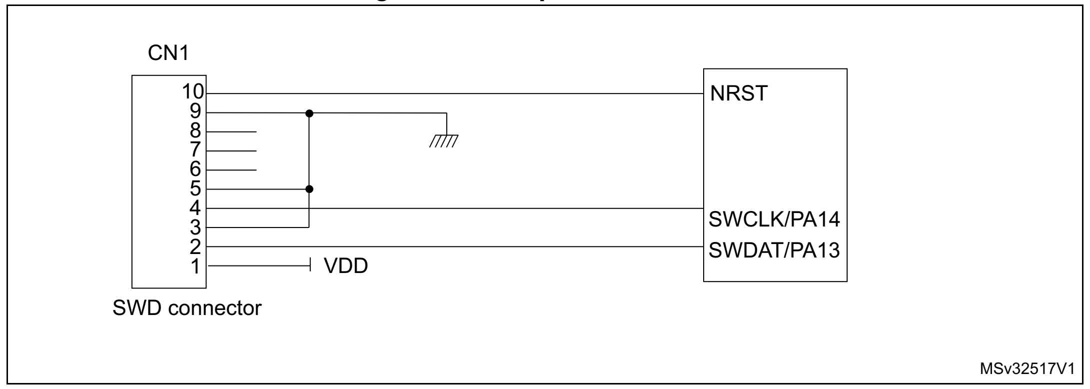
- SWDIO: alternate function pull-up
- SWCLK: alternate function pull-down

Having embedded pull-up and pull-down resistors removes the need to add external resistors.

4.3.4 SWD port connection with standard SWD connector

Figure 7 shows the connection between the STM32F0x0 and a standard SWD connector.

Figure 7. SWD port connection



4.3.2 SWD 引脚分配

复位（SYSRESETn 或 PORESETn）后，用于 SWD 的引脚被分配为专用引脚，调试器主机可以立即使用它们。

然而，MCU 提供了禁用 SWD 的可能性，从而释放相关引脚以供通用 I/O (GPIO) 使用。有关如何禁用 SWD 端口的更多详细信息，请参阅参考手册 STM32F030x4/6/8/C 和 STM32F070x6/B 高级基于 ARM® 的 32 位 MCU (RM0360) 中 I/O 引脚复用功能多路复用器和映射部分。

4.3.3 SWD 引脚上的内部上拉和下拉

一旦用户软件释放 SWD I/O，GPIO 控制器就会控制这些引脚。 GPIO 控制寄存器的复位状态将 I/O 置于等效状态：

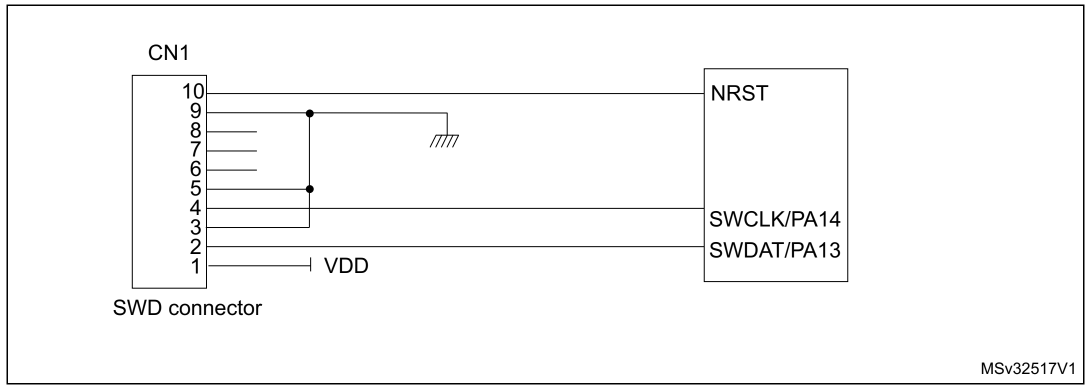
- SWDIO：复用功能上拉
- SWCLK：复用功能下拉

内置上拉和下拉电阻，无需添加外部电阻。

4.3.4 SWD 端口与标准 SWD 连接器的连接

图 7 显示了 STM32F0x0 和标准 SWD 连接器之间的连接 .

图 7. SWD 端口连接



5 Recommendations

5.1 Printed circuit board

For technical reasons, it is best to use a multilayer printed circuit board (PCB) with a separate layer dedicated to ground (V_{SS}) and another dedicated to the V_{DD} supply. This provides good decoupling and a good shielding effect. For many applications, economical reasons prohibit the use of this type of board. In this case, the major requirement is to ensure a good structure for ground and for the power supply.

5.2 Component position

A preliminary layout of the PCB must make separate circuits:

- High-current circuits
- Low-voltage circuits
- Digital component circuits
- Circuits separated according to their EMI contribution. This will reduce cross-coupling on the PCB that introduces noise.

5.3 Ground and power supply (V_{DD}, V_{DDA})

Every block (noisy, low-level sensitive, digital, etc.) should be grounded individually and all ground returns should be to a single point. Loops must be avoided or have a minimum area. In order to improve analog performance, the user must use separate supply sources for V_{DD} and V_{DDA}, and place the decoupling capacitors as close as possible to the device. The power supplies should be implemented close to the ground line to minimize the area of the supplies loop. This is due to the fact that the supply loop acts as an antenna, and is therefore the main transmitter and receiver of EMI. All component-free PCB areas must be filled with additional grounding to create a kind of shielding (especially when using single-layer PCBs).

5.4 Decoupling

All power supply and ground pins must be properly connected to the power supplies. These connections, including pads, tracks and vias should have as low an impedance as possible. This is typically achieved with thick track widths and, preferably, the use of dedicated power supply planes in multilayer PCBs.

In addition, each power supply pair should be decoupled with 100 nF filtering ceramic capacitor and a chemical capacitor of about 4.7 µF connected between the supply pins of the STM32F0x0 device. These capacitors need to be placed as close as possible to, or below, the appropriate pins on the underside of the PCB. Typical values are 10 nF to 100 nF, but exact values depend on the application needs. *Figure 8* shows the typical layout of such a V_{DD}/V_{SS} pair.

5 建议

5.1 印刷电路板

出于技术原因，最好使用多层印刷电路板 (PCB)，其中一层专用于接地 (VSS)，另一层专用于 VDD 电源。这提供了良好的去耦和良好的屏蔽效果。对于许多应用来说，经济原因禁止使用这种类型的板。在这种情况下，主要要求是确保良好的接地和电源结构。

5.2 元件位置

PCB 的初步布局必须制作单独的电路：

- 大电流电路
- 低压电路
- 数字元件电路
- 电路根据 EMI 贡献进行分离。这将减少 PCB 上引入噪声的交叉耦合。

5.3 地和电源（VDD 、 VDDA）

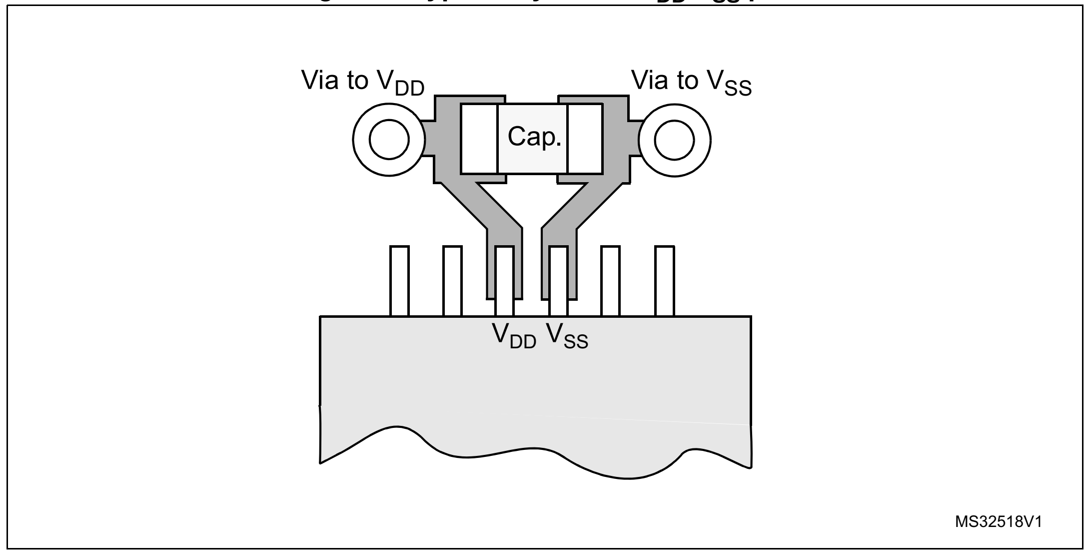
每个模块（噪声模块、低电平敏感模块、数字模块等）均应单独接地，并且所有接地回路均应连接至单点。必须避免环路或具有最小面积。为了提高模拟性能，用户必须使用独立的 VDD 和 VDDA 电源，并将去耦电容尽可能靠近器件放置。电源应靠近地线实施，以尽量减少电源环路的面积。这是因为电源环路充当天线，因此是 EMI 的主要发射器和接收器。所有无元件 PCB 区域都必须填充额外的接地以形成某种屏蔽（特别是在使用单层 PCB 时）。

5.4 解耦

所有电源和接地引脚必须正确连接到电源。这些连接，包括焊盘、走线和过孔，应具有尽可能低的阻抗。这通常通过较厚的走线宽度来实现，并且最好在多层 PCB 中使用专用电源层。

此外，每个电源对应通过连接在 STM32F0x0 器件的电源引脚之间的 100 nF 滤波陶瓷电容器和约 4.7 µF 的化学电容器进行去耦。这些电容器需要尽可能靠近 PCB 下侧的相应引脚或位于其下方。典型值为 10 nF 至 100 nF，但确切值取决于应用需求。图 8 显示了此类 VDD/VSS 对的典型布局。

Figure 8. Typical layout for V_{DD}/V_{SS} pair



5.5 Other signals

When designing an application, the EMC performance can be improved by closely studying:

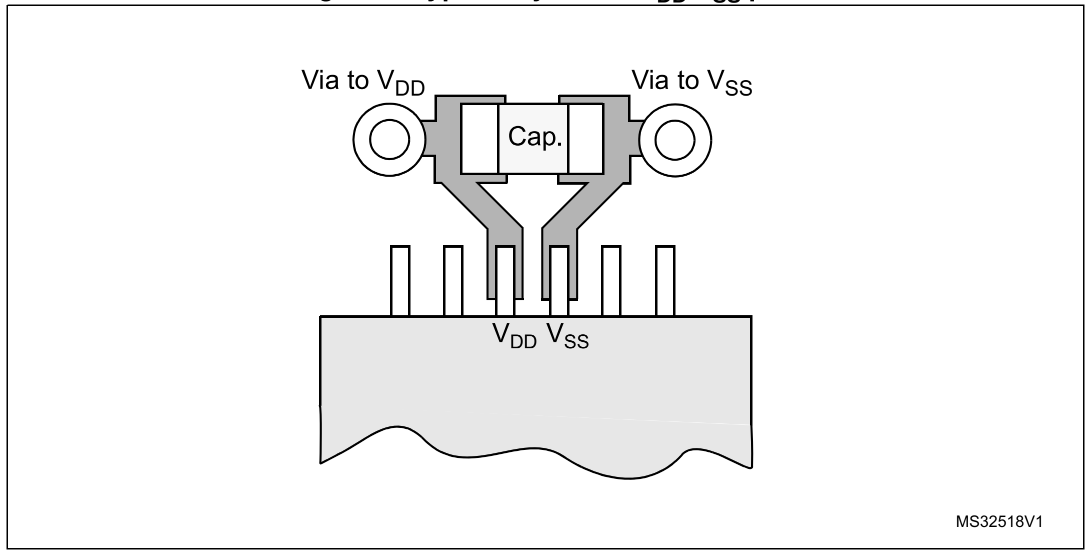
- Signals for which a temporary disturbance affects the running process permanently (such as interrupts and handshaking strobe signals, but not LED commands). For these signals, a surrounding ground trace, shorter lengths and the absence of noisy and sensitive traces nearby (crosstalk effect) improve EMC performance.
- Digital signals: the best possible electrical margin must be reached for the two logical states and slow Schmitt triggers are recommended to eliminate parasitic states.
- Noisy signals (clock, etc.)
- Sensitive signals (high impedance, etc.)

5.6 Unused I/Os and features

All microcontrollers are designed for a variety of applications and often a particular application does not use 100% of the MCU resources.

To increase EMC performance and avoid extra power consumption, unused clocks, counters or I/Os, should not be left free. I/Os should be connected to a fixed logic level of 0 or 1 by an external or internal pull-up or pull-down on the unused I/O pin. The other option is to configure GPIO as output mode using software. Unused features should be frozen or disabled, which is their default value.

Figure 8. Typical layout for V_{DD}/V_{SS} pair



5.5 其他信号

在设计应用时，可以通过仔细研究以下内容来提高 EMC 性能：

- 暂时干扰会永久影响运行过程的信号（例如中断和握手选通信号，但不包括 LED 命令）。对于这些信号，周围的接地走线、较短的长度以及附近没有噪声和敏感走线（串扰效应）可提高 EMC 性能。
- 数字信号：两个逻辑状态必须达到最佳的电气裕度，建议使用慢速施密特触发器来消除寄生状态。
- 噪声信号（时钟等）
- 敏感信号（高阻抗等）

5.6 未使用的 I/O 和功能

所有微控制器都是针对各种应用而设计的，通常特定应用不会使用 100% 的 MCU 资源。

为了提高 EMC 性能并避免额外的功耗，不应闲置未使用的时钟、计数器或 I/O。 I/O 应通过未使用的 I/O 引脚上的外部或内部上拉或下拉连接到固定逻辑电平 0 或 1。另一种选择是使用软件将 GPIO 配置为输出模式。未使用的功能应被冻结或禁用，这是它们的默认值。

6 Reference design

6.1 Description

The reference design shown in *Figure 9*, introduces the STM32F0x0, a highly integrated microcontroller running at 48 MHz, that combines the Cortex®-M0 32-bit RISC CPU core with 64 Kbytes of embedded Flash memory and 8 Kbytes of SRAM.

6.1.1 Clock

Two clock sources are used for the microcontroller:
- HSE: X1– 8 MHz crystal for the STM32F0x0 microcontroller
- LSE: X2– 32.768 kHz crystal for the embedded RTC

Refer to *Section 2: Clocks on page 11*.

6.1.2 Reset

The reset signal in *Figure 9* is active low. The reset sources include:
- Reset button (B1)
- Debugging tools via the connector CN1

Refer to *Section 1.2.2: System reset on page 9*.

6.1.3 Boot mode

The boot option is configured by setting BOOT0 through switch SW1 and option bit nBOOT1. Refer to *Section 3: Boot configuration on page 15*.

6.1.4 SWD interface

The reference design shows the connection between the STM32F0x0 and a standard SWD connector. Refer to *Section 4: Debug management on page 17*.

Note: It is recommended to connect the reset pin in order to be able to reset the application from the tool.

6.1.5 Power supply

Refer to *Section 1.1: Power supply schemes on page 6*.

6.1.6 Pinouts and pin description

Please refer to the STM32F0x0 datasheets available on *www.st.com* for the pinout information and pin description of each device.

6 参考设计

6.1 说明

图 9 所示的参考设计介绍了 STM32F0x0，这是一款运行频率为 48 MHz 的高度集成微控制器，它将 Cortex®-M0 32 位 RISC CPU 内核与 64 KB 嵌入式闪存和 8 KB SRAM 结合在一起。

6.1.1 时钟

微控制器使用两个时钟源：
- HSE：用于 STM32F0x0 微控制器的 X1– 8 MHz 晶振
- LSE：X2– 32.768 kHz 晶体，用于嵌入式 RTC

请参阅第 11 页第 2 节：时钟。

6.1.2 复位

图 9 中的复位信号为低电平有效。复位源包括：
- 重置按钮（ B1 ）
- 通过连接器 CN1 调试工具

请参阅第 9 页第 1.2.2 节：系统重置。

6.1.3 启动模式

通过开关 SW1 和选项位 nBOOT1 设置 BOOT0 来配置引导选项。请参阅第 3 节：第 15 页的引导配置。

6.1.4 SWD 接口

该参考设计显示了 STM32F0x0 和标准 SWD 连接器之间的连接。请参阅第 17 页第 4 节：调试管理。

笔记： 建议连接重置引脚，以便能够从工具重置应用程序。

6.1.5 电源

请参阅第 1.1 节：第 6 页的电源方案。

6.1.6 引脚排列和引脚描述

请参阅 www.st.com 上提供的 STM32F0x0 数据表，了解每个器件的引脚排列信息和引脚描述。

6.2 Component references

Table 5. Mandatory components

Component	Reference	Value	Quantity	Comments
Microcontroller	U1	STM32F030R8T6	1	64-pin package.
Capacitor	C1/C2	100 nF	2	Ceramic capacitors (decoupling capacitors).
Capacitor	C3	10 nF	1	Ceramic capacitor (decoupling capacitor).
Capacitor	C5	1 µF	1	Used for VDDA.
Capacitor	C6	4.7 µF	1	Used for VDD.

Table 6. Optional components

Component	Reference	Value	Quantity	Comments
Resistor	R1	390 Ω	1	Used for HSE: the value depends on the crystal characteristics. This value is given only as a typical example.
Resistor	R2	10 KΩ	1	Used for BOOT0 pin.
Capacitor	C4	100 nF	1	Ceramic capacitor for RESET button.
Capacitor	C7/C8	10 pF	2	Used for LSE: the value depends on the crystal characteristics.
Capacitor	C9/C10	20 pF	2	Used for HSE: the value depends on the crystal characteristics.
Quartz	X1	8 MHz	1	Used for HSE.
Quartz	X2	32 kHz	1	Used for LSE.
Switch	SW1	-	1	Used to select the correct boot mode.
Push-button	B1	-	1	Used as reset button.
SWD connector	CN1	FTSH-105-01-L-DV	1	Used for program/debug of the MCU.

6.2 组件参考

表 5. 强制组件

Component	Reference	Value	Quantity	Comments
Microcontroller	U1	STM32F030R8T6	1	64-pin package.
Capacitor	C1/C2	100 nF	2	Ceramic capacitors (decoupling capacitors).
Capacitor	C3	10 nF	1	Ceramic capacitor (decoupling capacitor).
Capacitor	C5	1 µF	1	Used for VDDA.
Capacitor	C6	4.7 µF	1	Used for VDD.

表 6. 可选组件

Component	Reference	Value	Quantity	Comments
Resistor	R1	390 Ω	1	Used for HSE: the value depends on the crystal characteristics. This value is given only as a typical example.
Resistor	R2	10 KΩ	1	Used for BOOT0 pin.
Capacitor	C4	100 nF	1	Ceramic capacitor for RESET button.
Capacitor	C7/C8	10 pF	2	Used for LSE: the value depends on the crystal characteristics.
Capacitor	C9/C10	20 pF	2	Used for HSE: the value depends on the crystal characteristics.
Quartz	X1	8 MHz	1	Used for HSE.
Quartz	X2	32 kHz	1	Used for LSE.
Switch	SW1	-	1	Used to select the correct boot mode.
Push-button	B1	-	1	Used as reset button.
SWD connector	CN1	FTSH-105-01-L-DV	1	Used for program/debug of the MCU.

Figure 9. STM32F030 microcontroller reference schematic

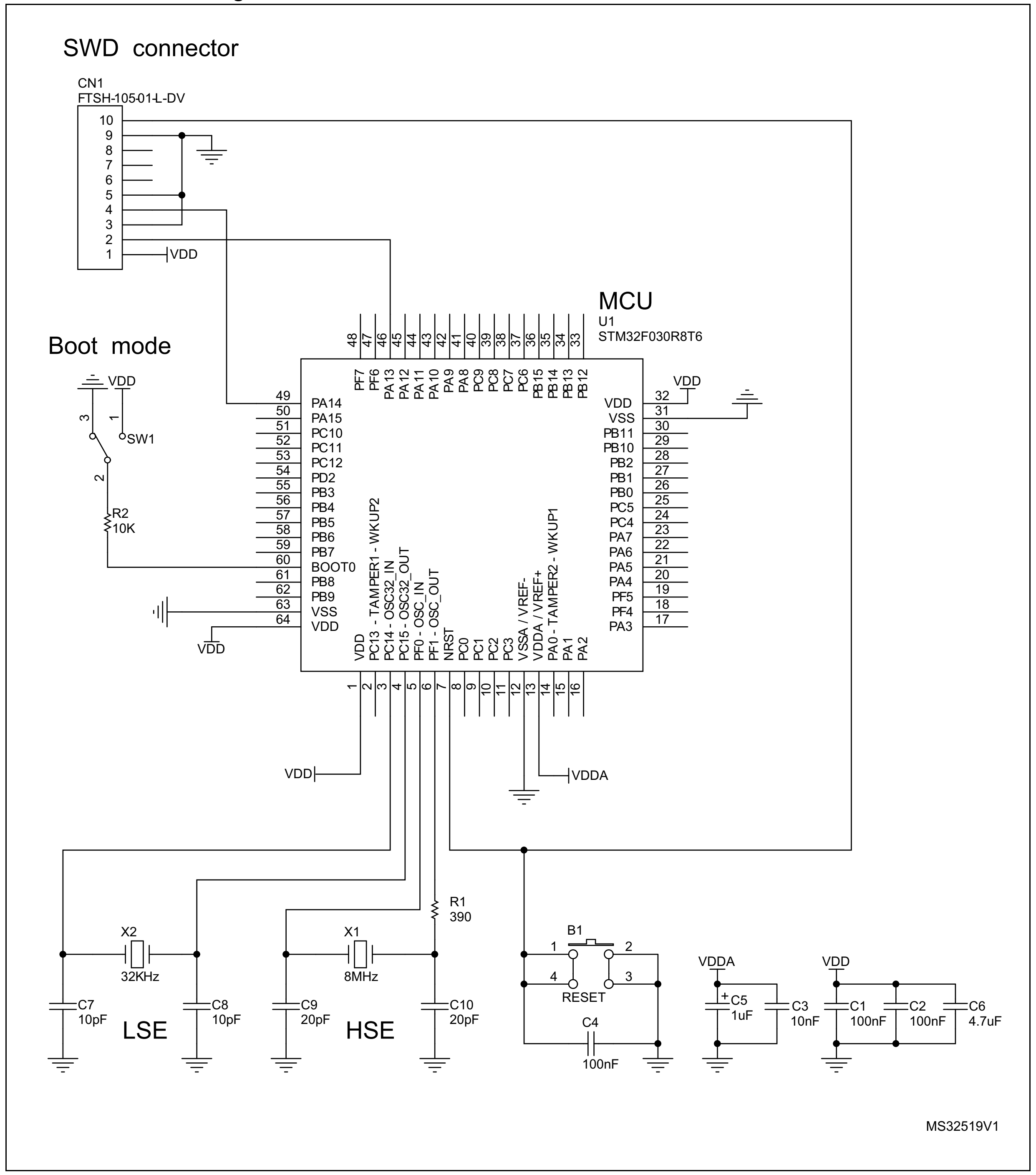
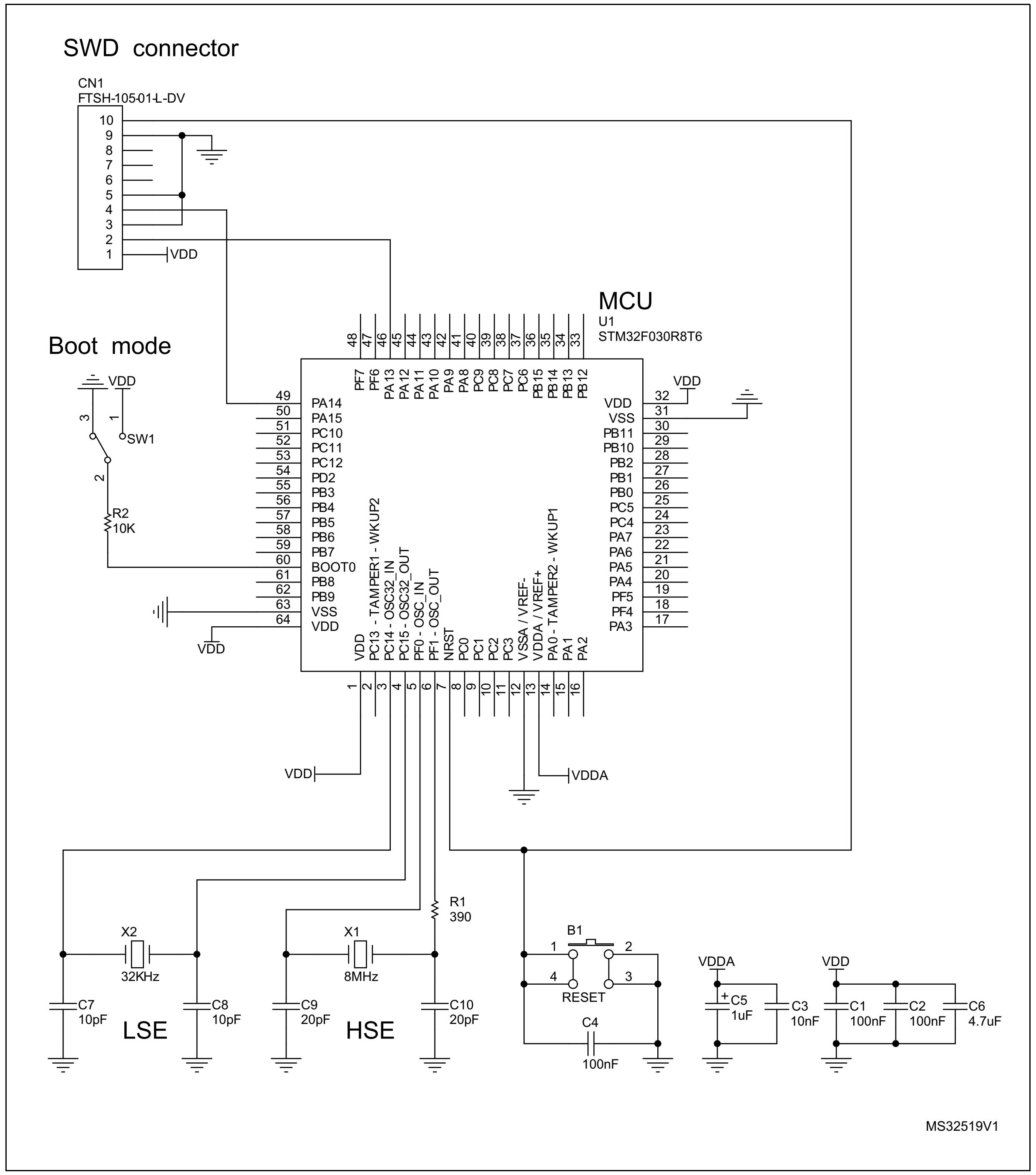


图 9. STM32F030 微控制器参考原理图



7 Hardware migration from STM32F1 series to STM32F0x0 devices

The entry-level STM32F030 and general-purpose STM32F1xxx families are pin-to-pin compatible. All peripherals shares the same pins in the two families, but there are some minor differences between packages.

The transition from the STM32F1 series to the STM32F030 devices is simple as only a few pins are impacted. The impacted pins are shown in bold in *Table 7*.

Table 7. STM32F1 and STM32F030 series pinout differences(1)

Package		STM32F1 series	STM32F030 devices
LQFP64	LQFP48	Pinout	Pinout
1	1	**VBAT**	**VDD**
5	5	**PD0** - OSC_IN	**PF0** - OSC_IN
6	6	**PD1** - OSC_OUT	**PF1** - OSC_OUT
18	-	**VSS_4**	**PF4**
19	-	**VDD_4**	**PF5**
28	20	**BOOT1** - PB2	PB2
47	35	**VSS_2**	**PF6**
48	36	**VDD_2**	**PF7**

1. Highlighted in **bold** the impacted pins of the transition between STM32F1 series to STM32F030 devices.

7 从 STM32F1 系列到 STM32F0x0 器件的硬件迁移

入门级 STM32F030 和通用 STM32F1xxx 系列引脚兼容。这两个系列中的所有外设共享相同的引脚，但封装之间存在一些细微差别。

从 STM32F1 系列到 STM32F030 器件的过渡很简单，因为只有少数引脚受到影响。受影响的引脚在表 7 中以粗体显示。

表 7. STM32F1 和 STM32F030 系列引脚排列差异 (1)

Package		STM32F1 series	STM32F030 devices
LQFP64	LQFP48	Pinout	Pinout
1	1	**VBAT**	**VDD**
5	5	**PD0** - OSC_IN	**PF0** - OSC_IN
6	6	**PD1** - OSC_OUT	**PF1** - OSC_OUT
18	-	**VSS_4**	**PF4**
19	-	**VDD_4**	**PF5**
28	20	**BOOT1** - PB2	PB2
47	35	**VSS_2**	**PF6**
48	36	**VDD_2**	**PF7**

1. Highlighted in **bold** the impacted pins of the transition between STM32F1 series to STM32F030 devices.

8 Revision history

Table 8. Document revision history

Date	Revision	Changes
10-Nov-2014	1	Initial release.
01-Apr-2016	2	Added the support for STM32F070xx along the document.

8 修订历史

表 8. 文档修订历史记录

Date	Revision	Changes
10-Nov-2014	1	Initial release.
01-Apr-2016	2	Added the support for STM32F070xx along the document.

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